

Unit 1, Unit 2, and Unit 3 (3.1 - 3.3) Practice

Section 1: Multiple Choice Questions (No calculator is allowed for these questions.)

1. The table below shows values of a function $f(x)$ at constant intervals of x .

x	0	2	4	6
$f(x)$	5	12	23	38

Based on the successive differences, which of the following describes the behavior of $f(x)$?

- (A) $f(x)$ is concave up because the first differences are increasing.
- (B) $f(x)$ is concave down because the first differences are decreasing.
- (C) $f(x)$ is linear because the second differences are constant.
- (D) $f(x)$ is concave down because the second differences are negative.
2. A geometric sequence g_n has terms $g_2 = 12$ and $g_5 = 96$. Assuming the common ratio r is a real number, what is the value of g_7 ?
- (A) 150
- (B) 192
- (C) 384
- (D) 768
3. The function $f(t)$ models a periodic phenomenon. The graph of f reaches its maximum value of 15 at $t = 2$. The very next maximum value occurs at $t = 10$. What is the value of $f(34)$?
- (A) 2
- (B) 8
- (C) 10
- (D) 15
4. Let θ be an angle in standard position. If $\sin \theta = \frac{5}{13}$ and $\tan \theta < 0$, what is the value of $\cos \theta$?

(A) $\frac{12}{13}$

(B) $\frac{5}{12}$

(C) $-\frac{5}{12}$

(D) $-\frac{12}{13}$

5. Which of the following expressions is equivalent to the exponential function

$$h(x) = 3 \cdot 4^{x-0.5}?$$

(A) $h(x) = 6 \cdot 4^x$

(B) $h(x) = 1.5 \cdot 4^x$

(C) $h(x) = 3 \cdot 2^x$

(D) $h(x) = 12^x \cdot 12^{-0.5}$

6. Point $P(x, y)$ lies on the unit circle and corresponds to an angle θ in standard position. If

$$x = -0.6 \text{ and } y > 0, \text{ what is the value of } \sin(\theta + \pi)?$$

(A) 0.8

(B) 0.6

(C) -0.6

(D) -0.8

7. An exponential function $E(x)$ has an Average Rate of Change (AROC) of 20 over the interval $[0, 3]$. Its AROC over the interval $[3, 6]$ is 100. What is the AROC of $E(x)$ over the interval $[6, 9]$?

(A) 180

(B) 300

(C) 500

(D) 800

8. Evaluate the exact value of $\tan\left(\frac{11\pi}{6}\right)$.

(A) $\sqrt{3}$

(B) $\frac{\sqrt{3}}{3}$

(C) $-\frac{\sqrt{3}}{3}$

(D) $-\sqrt{3}$ **Section 2: Free Response Questions (Show all your work clearly.)****FRQ 1**

A biologist is studying the population of a certain bacteria culture. The population $P(t)$ is recorded every 3 hours, as shown in the table below.

t (hours)	0	3	6	9
$P(t)$ (thousands)	100	50	25	12.5

- (a) Calculate the Average Rate of Change (AROC) of the bacteria population over the interval $[0, 3]$ and the interval $[3, 6]$. Include units in your answer.
- (b) Based on your calculations in part (a), is the graph of $P(t)$ concave up or concave down? Justify your reasoning using the behavior of the AROC.
- (c) The population can be modeled exactly by an exponential function in the form $P(t) = a \cdot b^t$. Find the exact values of the constants a and b .
- (d) Using the model $P(t)$ found in part (c), determine the exact time t (in hours) when the population of the bacteria culture will reach 6.25 thousand. Show your algebraic steps.

FRQ 2

A weight attached to a mechanical spring bounces up and down. Its height above the ground, $h(t)$, measured in centimeters, is modeled by a periodic function where t is the time in seconds.

The weight is at its maximum height of 18 cm at $t = 1$ second. It reaches its very next maximum height at $t = 5$ seconds. The minimum height of the weight during its cycle is 6 cm.

- (a) Determine the period of $h(t)$ and explain the mathematical meaning of the period in the context of this scenario.
- (b) Find the height of the weight at $t = 29$ seconds. Show the mathematical reasoning that leads to your answer.
- (c) At what exact time t , where $0 < t < 5$, does the weight reach its minimum height of 6 cm? Justify your reasoning.
- (d) A new function $d(t) = h(t) - 12$ is created to represent the displacement of the weight from its resting position. What are the maximum and minimum values of $d(t)$?

FRQ 3

Consider two angles, α and β , both in standard position on the coordinate plane.

The terminal side of angle α passes through the point $A(-2, \sqrt{5})$. Angle β is known to satisfy the following conditions: $\sin \beta = -\frac{1}{2}$ and $\cos \beta > 0$.

(a) Find the exact value of the radius r for point A , and use it to determine the exact values of $\sin \alpha$ and $\cos \alpha$.

(b) Determine which quadrant contains the terminal side of angle β . Justify your answer using the given conditions.

(c) Given that $0 \leq \beta < 2\pi$, determine the exact measure of angle β in radians.

(d) Evaluate the exact value of $\tan(\beta + \pi)$. Show your reasoning.